JEDIDIAH ROSEMAN

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EDUCATION

University of Michigan *Ann Arbor, Michigan*
Bachelor of Science in Electrical Engineering | Major GPA: 3.91 *Expected May 2028*
Relevant Coursework: Intro Circuits, EE System Design, Electromagnetics, Multivariable Calculus, Differential Equations; currently taking (Fall 2026): Programming & Data Structures, Signals & Systems, Semiconductor Devices, and Probabilistic Methods in Engineering

TECHNICAL PROJECTS

Autonomous Warehouse Inventory Robot *Atlanta, Georgia*
Independent Hardware Project | ROS2, Raspberry Pi, Arduino, RPLIDAR *Summer 2026 - Present*

- Building a camera-based barcode-scanning robot, now driving, to count $4.4M of inventory across 1,598 SKUs at Payless Decor, replacing a two-person, 7-14 workday manual count on ladders and forklifts; designed to run overnight so counts go from quarterly to monthly.
- Caught that the vendor encoder spec assumed 4× quadrature decoding, which the ATmega328P's two hardware interrupts could not support; redesigned for 2× decoding at 0.62 mm/count and wrote firmware to correct odometry drift across all four drive motors.
- Sized the power system around a 12.8 A worst-case four-motor stall current, cutting the main fuse rating from 40 A to 20-25 A so overcurrent protection trips ahead of the BMS threshold and a fault clears with a two-dollar part instead of a manual battery pack reset.
- Wrote the Raspberry Pi-to-Arduino UART firmware with framed motor commands, a 20 Hz odometry stream, and a 500 ms watchdog that halts drive on link loss; diagnosed and cleared 5 hardware and spec defects during bring-up, including misaligned motor inserts.

Sensor-Based Attendance Counter *Ann Arbor, Michigan*
5-Person Team Project | Arduino, C++, Ultrasonic Sensors, Onshape *Spring 2025*

- Built a real-time bidirectional people counter using dual ultrasonic sensors (chosen over passive infrared and camera-based options) in a battery-powered, 3D-printed door-frame unit built for under $40, detecting entry vs. exit by order of occlusion.
- Achieved 98% counting accuracy across repeated lecture hall trials and owned all hardware integration, including sensor wiring, power routing with an inline switch to prevent idle battery drain, pull-down resistors on all inputs, and sensor testing across distance and angle.

Midtown Electric Vehicle Rally *Atlanta, Georgia*
Co-Founder and Head of Logistics *2022 - 2024*

- Engineered the full wiring system and prototyped PCBs for a brushless DC motor drivetrain, collaborating on electrical and mechanical integration across a 20+ member team to field a 65 MPH vehicle that placed in national 2-hour endurance races over two seasons.
- Secured $25K in sponsorship from Mercedes-Benz and Rivian via executive outreach to launch Atlanta's first high school EV club.

LEADERSHIP

Michigan Defense Technology (MDT) *Ann Arbor, Michigan*
Co-President and Founder *2025 - Present*

- Led outreach and scoping for 3 sponsored projects, turning requirements into deliverables for 6-10 person teams: an AUV research platform with Booz Allen, a drone swarm kit for AG3 tested in GPS-denied environments, and a wind tunnel data pipeline for North Wind.
- Co-founded and scaled U-M's first defense tech organization to 100+ applicants at an 18% acceptance rate while defining the strategic roadmap for technical research, operations, recruitment, and partnerships with Palantir, Anduril, Scale AI, Shield AI, and Antares.

WORK EXPERIENCE

First Financial Bank USA *Atlanta, Georgia*
Product Manager, Contracted Software Project *Summer 2026*

- Proposed the product and hired 6 interns across 6 workstreams spanning frontend, backend, quantitative modeling, and compliance.
- Led the team through a 13-week build of a full-stack fintech MVP targeting $50M in retail IRA deposits, clearing all 3 executive gate reviews on schedule and delivering a production-ready prototype with full technical documentation for an external software firm.

Additional Experience: Investment Analyst, Service Provider Capital (2025); VP of Consulting, Ascend Consulting Group (2024 - Present)

SKILLS & HONORS

Hardware: Circuit design, PCB schematic capture and layout, fuse sizing, soldering and crimping, brushless DC motor drivetrains, hardware bring-up and fault isolation, sensor integration (LIDAR, ultrasonic, IMU, camera), UART and I2C, oscilloscope, multimeter, 3D printing
Software & Tools: C++, Python, MATLAB, LTspice, Altium, KiCad, ROS2, Arduino, Raspberry Pi, Onshape (CAD), Linux, Git, TypeScript, React
Honors: Harvard National Debate Tournament Quarter Finalist (Top 100 in US); MIT Inspirit AI, built a CNN chest X-ray pneumonia classifier
Hobbies: App Development, Video Production, Backpacking, Basketball, Golf, Music (Vinyl)